AGRICULTURAL MATERIALS

Determination of Plant Hormones in Fertilizers by High-Performance Liquid Chromatography with Photodiode Array Detection: Method Development and Single-Laboratory Validation

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A simple and reliable high-performance liquid chromatographic method that uses photodiode array detection was developed for the simultaneous determination of 12 native and synthetic plant hormones, i.e., plant growth regulators (PGRs), in fertilizers, such as 1-naphthol, 2,4-dichlorophenoxyacetic acid, 4-(2,4-dichlorophenoxy)butyric acid, 4-chlorophenoxyacetic acid, indole-3-acetic acid, 4-(3-indolyl)butyric acid, dichlorprop, (4-chloro-2-methylphenoxy)acetic acid, α-naphthaleneacetic acid, 1-naphthalene-acetamide, β-naphthoxyacetic acid, and thidiazuron. The method was experimentally validated for routine regulatory application, and the following analytical parameters were assessed for all PGRs studied: linearity; specificity; precision (relative standard deviation) and accuracy, both measured at 3 concentration levels (0.1, 0.05, and 0.01%, w/w); ruggedness; limit of detection; and limit of quantification. Results were satisfactory for all method validation parameters tested and for all PGRs studied, demonstrating the suitability of the method for the determination of PGRs in fertilizers. The uncertainty of measurement was also estimated at 3 concentration levels for all PGRs by using the approach of the International Organization for Standardization, described in its *Guide to the Expression of Uncertainty in Measurement*. The method was applied to 20 samples of liquid fertilizer with declared biostimulant properties.

Plant hormones, or plant growth regulators (PGRs), are internally secreted chemicals in plants that regulate the growth of plants. According to a standard definition, plant hormones are signal molecules produced at specific locations in very low concentrations and cause altered processes in target cells at other locations. The concentration of hormones required for plant response is very low (10^{-6}–10^{-5} M), compared with the requirements of minerals and vitamins for plants. It is accepted that there are 5 major classes of plant hormones: auxins, cytokinins, ethylene, gibberellins, and abscisic acid (1, 2). In addition to natural phytohormones, there are less expensive and more stable synthetic analogs on the market for use in horticulture. They can influence the life processes of plants, destroy the undesired parts of plants, or control or prevent the undesired growth of plants. In the European Union (EU) the use of PGRs for commercial purposes is subject to the legislation in Council Directive 91/414/EEC. All EU member states apply the same evaluation procedures and authorization criteria in placing a plant protection product on the market.

The Italian Legislative Decree 217/2006 concerning new regulations for fertilizers originates from European Regulation 2003/2003/EC and deals with the official control and marketing of fertilizers. An innovation in this decree is the introduction of fertilizer products described as "biostimulants," specifying that the stimulant effect is limited to the presence of natural compounds derived from algae, humic substances, amino acids, potassium silicates, and salicylic acid (3). These substances should improve effectiveness of the nutrient absorption and promote and stimulate plant growth. Therefore, new analytical methods are needed to verify both the absence of added synthetic hormones and the presence of effective biostimulant activity (3). Many methods have been reported for the determination of plant hormones in different matrixes (4–7), but no researchers have reported the use of plant hormones to improve the biostimulant effect of common fertilizers.

This paper reports a reliable analytical method for the simultaneous routine determination of several PGRs in fertilizers. The method combines high-performance liquid chromatographic (HPLC) separation with photodiode array detection (PDA) to detect and quantify native and synthetic

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PGRs. The compounds investigated were 4-(3-indolyl)butyric acid (IBA), 1-naphthaleneacetamide (NAD), α-naphthaleneacetic acid (NAA), β-naphthoxyacetic acid (β-NOA), 4-chlorophenoxyacetic acid (4-CPA), (4-chloro-2-methylphenoxy)acetic acid (MCPA), 2,4-dichlorophenoxyacetic acid (2,4-D), 4-(2,4-dichlorophenoxy)butyric acid (2,4-DB), indole-3-acetic acid (IAA), 1-naphthol, 1-phenyl-3-(1,2,3-thiadiazol-5-yl)urea (thidiazuron), and 2-(2,4-dichlorophenoxy)propanoic acid (dichlorprop). In order to demonstrate the suitability of the method for routine regulatory purposes, the method was validated for the determination of PGRs in fertilizers, and the statistical parameters are discussed.

Experimental

Chemicals

Acetic acid was purchased from Fluka (Milan, Italy). NAD (purity 98%), 2,4-DB (95%), and 1-naphthol (100%) were from Sigma-Aldrich (Milan, Italy), and NAA (95%), β-NOA (95%), 4-CPA (98%), MCPA (97%), 2,4-D (95%), IAA (97%), IBA (95%), diclorprop (95%), and thidiazuron (99%) were supplied by Fluka. HPLC grade acetonitrile and methanol were supplied by Romil (Milan, Italy). Distilled water was purified at 18.2 MΩ·cm with a Millipore Milli-Q ULTRA (Vimodrone, Italy) purification system.

Preparation and Storage of PGR Standards

Accurately weighed 50.0 mg portions of each compound were placed in separate 50 mL volumetric flasks, and the contents of the flasks were diluted to volume with methanol to obtain stock standard solutions of the PGRs. Aliquots of each solution were further diluted to obtain 2 working standard mixtures of the PGRs. The PGRs were distributed as follows:

Mixture 1: NAD, 2,4-D, β-NOA, IAA, and IBA; Mixture 2: 1-naphthol, NAA, 4-CPA, MCPA, 2,4-DB, diclorprop, and thidiazuron. The reference PGR stock standard solutions were stored at 4°C; under these conditions, they were stable for 3 months.

Sample Preparation

The analyzed matrixes were liquid commercial samples with declared biostimulant properties and were processed as follows: 500.0 mg homogenized sample was weighed and dissolved in methanol; the resulting solution was spiked with known amounts of the working standard mixtures of the PGRs and diluted to volume with methanol in a volumetric flask. The solution was sonicated in an ultrasonic bath at 20°C and filtered through a 0.45 μm polytetrafluoroethylene (PTFE) filter (Labservice Analytica, Bologna, Italy) before HPLC analysis.

HPLC/PDA Conditions for Determination of PGRs

Separation was obtained with a Thermo Finnigan Surveyor (Thermo Fisher Scientific, Rodano, Milan, Italy) HPLC system equipped with an autosampler and a variable wavelength spectrophotometer. A Synergi Hydro-RP polar encapped C_{18} column, 250 × 4.60 mm (4 μm; Phenomenex, Anzola Emilia [BO], Italy) was used and the column temperature was maintained at 30°C. The range of wavelengths examined by the photodiode array detector was 200–360 nm, and for quantitation the chromatograms were recorded at 280 nm. A flow rate of 1.0 mL/min for 30 min was used. The composition of the mobile phase was (0.8% acetic acid) water–acetonitrile (55 + 45, v/v). The injection volume was 20 μL.

Statistical Data and Method Validation

Specificity.—The specificity of the analytical method for the determination of PGRs was confirmed by comparing fortified fertilizers with negative controls. Three types of matrixes were screened for use as the control: liquid extracts from algae, liquid organic fertilizers, and liquid extracts from vegetal tissue containing amino acids and polysaccharides. Two samples of each matrix were analyzed. A liquid organic fertilizer was chosen as the negative control for the determination of method validation parameters.

Linearity.—The solutions for the HPLC/PDA calibration curves were obtained by diluting 6 independent stock solutions of the 2 mixtures of PGRs over the range of about 0.5–25 mg/L, making 4 replicates for each concentration. The peak area integration was performed at 280 nm, and the relationship with concentration was determined by using linear regression. The regression equation with slope, intercept, and coefficient of correlation (r^2) was evaluated for each PGR.

System precision.—Instrumental precision was checked by repeated scanning (n = 7) of the same solution of standard PGRs, and the relative standard deviation (RSD_{INSTR}) was calculated for each analyte as follows:

$$RSD_{INSTR}, \% = (SD \times 100)/C$$

where RSD = relative standard deviation, SD = standard deviation, and C = mean of the calculated concentration.

Limit of detection (LOD) and limit of quantification (LOQ).—At least 7 replicates of blank samples (negative controls) spiked with PGRs at a level close to the LOD were prepared, and the SD values were determined. The LOD and the LOQ were obtained as follows:

$$LOD_{INSTR}, mg/L = 3 \times SD$$
$$LOQ_{INSTR}, mg/L = 10 \times SD$$

The LOD_{METH} (%, w/w) and LOQ_{METH} (%, w/w) were calculated by considering the weight of the sample and the final volume of the method protocol.

Method precision (repeatability).—Method precision was evaluated by analyzing 10 independent solutions of fortified fertilizer at 3 different concentration levels (0.1, 0.05, and 0.01%, w/w). The intraday precision was calculated for each concentration level as the RSD (%).

Accuracy (recovery).—The accuracy of the method was ascertained by spiking negative controls with a known concentration of PGRs. The spiking was done 10 times at

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Table 1. Design of Youden ruggedness trial and calculations

Factors	Level 1	Level 2
Weight of sample, mg	750 (A)	200 (a)
Volume of solvent, mL	40 (B)	20 (b)
Sonication time, min	22 (C)	10 (c)
Matrix type	Proteic vegetal tissue extract (D)	Algae extract (d)
Solvent composition	Methanol–water (70 + 30, v/v) (E)	Methanol (e)
Ultrasonic bath temperature, °C	40 (F)	30 (f)
Column temperature, °C	35 (G)	25 (g)
Experiment No.		
1	ABCDEFG	X1[a]
2	ABcDefg	X2[a]
3	AbCdEfg	X3[a]
4	AbcdeFG	X4[a]
5	aBCdeFg	X5[a]
6	aBcdEfG	X6[a]
7	abCDefG	X7[a]
8	abcDEFg	X8[a]
Effect of A and a ($V_{A/a}$)	[(X1 + X2 + X3 + X4)/4] – [(X5 + X6 + X7 + X8)/4]	
Effect of B and b ($V_{B/b}$)	[(X1 + X2 + X5 + X6)/4] – [(X3 + X4 + X7 + X8)/4]	
Effect of C and c ($V_{C/c}$)	[(X1 + X3 + X5 + X7)/4] – [(X2 + X4 + X6 + X8)/4]	
Effect of D and d ($V_{D/d}$)	[(X1 + X2 + X7 + X8)/4] – [(X3 + X4 + X5 + X6)/4]	
Effect of E and e ($V_{E/e}$)	[(X1 + X3 + X6 + X8)/4] – [(X2 + X4 + X5 + X7)/4]	
Effect of F and f ($V_{F/f}$)	[(X1 + X4 + X5 + X8)/4] – [(X2 + X3 + X6 + X7)/4]	
Effect of G and g ($V_{G/g}$)	[(X1 + X4 + X6 + X7)/4] – [(X2 + X3 + X5 + X8)/4]	

[a] Experimental concentration found for each experiment (n = 2); spiked with about 0.1%, w/w, of each PGR.

3 different concentration levels (0.1, 0.05, and 0.01%, w/w) of the estimated PGR concentrations in the sample). The average percent recovery (E) at each concentration level was calculated as follows:

$$E, \% = (C \times 100)/TC$$

where E = accuracy, C = mean of the calculated concentrations, and TC = theoretical concentration.

Youden ruggedness testing.—A ruggedness trial was performed for 7 factors potentially affecting the quantitative results: weight of sample, volume of extraction solvent, polarity of extraction solvent, ultrasonic bath temperature, column temperature, extraction time, and matrix typology. Table 1 reports the design of the ruggedness trial and the related calculations. The results were compared with the repeatability data obtained during the validation process for the corresponding fortified concentration level (0.1%, w/w). To consider the method rugged, it must meet the following requirement (8):

$$SD_{Ru}^2/SD_M^2 < F_{v = 7.9\ \alpha = 0.05}$$

where SD_{Ru} is the ruggedness test SD; SD_M is the method SD (spiking concentration, 0.1%, w/w; n = 10); F is the F-test; v is the number of degrees of freedom; and α is the level of significance.

Estimation of the effect of each parameter on method performance was evaluated as follows:

$$|V_{X/x}| \leq t_{p = 0.95;\ v = 9}\, SD_M/\sqrt{2}$$

where $|V_{X/x}|$ is the effect of X and x; X is the parameter considered; v is the number of degrees of freedom; SD_M is the method SD (spiking concentration, 0.1%, w/w; n = 10); t is the Student's t-value; and p is the level of confidence.

Quantification of uncertainty sources.—The measurand is the concentration of PGR in the fertilizer (%X), calculated as follows:

$$\%X = Cx \cdot V \cdot \frac{1}{M} \cdot E \cdot 10^{-3}$$

where Cx is the PGR concentration obtained by using the calibration curve (mg/L), V is the volume (L), M is the sample weight (g), E is the recovery (%), and 10^{-3} is the conversion factor (mg to g). Because no reference material certified for

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this standardized method is available, overall control of bias is related to the control of method parameters influencing the result.

PGR concentration, Cx.—The amount of PGR is calculated by using a calibration curve. For this purpose, 6 calibration standards were prepared from a reference standard. The relative uncertainty associated with the linear least-squares fitting procedure is calculated as follows:

$$\left[\dot{u}_B\left(\overline{C}x\right)\right]=\sqrt{\frac{\left[s(r)\right]^2}{b^2}\times\left[\frac{1}{n_1}+\frac{1}{m}+\frac{\left(y-\overline{y}_i\right)^2}{b^2\sum\left(X_i-\overline{X}_i\right)^2}\right]\times\frac{1}{n}\times\frac{1}{\overline{C}x}}$$

where $\overline{C}x$ is the mean value of Cx, b is the slope, n is the number of measurements to determine X, m is the number of measurements for the calibration, n_1 is the number of measurements for each calibration point, X_i is the determined concentration of the PGR solution, $\overline{X}_i$ is the mean value of the different calibration standards, i is the index for the number of the calibration standards, y is the area of the calibration point at the level studied, $\overline{y}_i$ is the mean value of the areas of the calibration standards over the calibration range,

$$\left[s(r)\right]=\sqrt{\frac{\sum\left(y_i-\hat{y}_i\right)^2}{n-2}}$$

$\hat{y}_i$ is the area value calculated by using the calibration curve and y_i is the experimental area value.

Mass, M.—The uncertainty arises from the balance linearity uncertainty. It is calculated by using a 7-point calibration curve purchased with the balance (resolution, 0.1 mg) and the linear least-squares fitting procedure.

Purity, P.—The purities of the PGRs are stated in the supplier's catalog to be within the limits of 95 and 100%. There is no other uncertainty source if the drying procedure is performed according to the supplier's specifications. Therefore, this relative uncertainty is assumed to have a rectangular distribution, and the standard uncertainty is as follows:

$$\left[\dot{u}_B(P)\right]=\frac{(100-P)}{100}\times\frac{1}{\sqrt{3}}$$

where P is the purity of standard PGR (%).

Recovery, E.—The bias of the analytical procedure was investigated during the in-house validation study by using spiked samples ($n = 20$). The relative standard uncertainty was calculated as the SD of the mean:

$$\left[\dot{u}_B\left(\overline{E}\right)\right]=\sqrt{\left[\frac{SD}{\sqrt{n}}\times\frac{1}{\overline{E}}\right]^2+\left[\dot{u}_B(ST)\right]^2}$$

where n is the number of measurements taken to determine $\overline{E}$, SD is the standard deviation of E, and $\dot{u}_B(ST)$ is the relative uncertainty of the PGR standard solution (as the sum of purity, dilution, and weight).

A significance test is used to determine whether the mean recovery is significantly different from 100%. The test statistic t is calculated by using the following equation:

$$t=\frac{\left|100-\overline{E}\right|}{u\left(\overline{E}\right)}$$

This value is compared with the 2-tailed critical value t_{crit}, for n–1 degrees of freedom at the 95% confidence level. If t is greater than or equal to the critical value t_{crit}, then E is significantly different from 100%.

$$t \geq t_{crit;\,n}$$

Volume, V.—The volume of the solution is subject to 3 sources of uncertainty: calibration, repeatability, and temperature effects. (*1*) *Calibration*.—The uncertainty in the certified internal volume of the flask and of the pipets. For example, the manufacturer gives a volume of 50 ± 0.06 mL (V ± a) for the flask, when measured at a temperature of 20°C. Because the value of the uncertainty is given without a confidence level or distribution information, an assumption is necessary. In this work, the standard uncertainty is calculated by assuming a rectangular distribution.

$$\left[\dot{u}_B(V)\right]=\frac{a}{\sqrt{3}}\times\frac{1}{V}$$

In the same way, the volumes of the pipets used to prepare the solutions of the PGRs at different levels are calculated by assuming a rectagular distribution. The contributions due to the dilution operations performed for each concentration level are calculated separately and combined to give the standard uncertainty of the volume. (*2*) *Repeatability*.—The variation in filling the flask to volume. The uncertainty due to variations in filling is considered in the repeatability experiments. (*3*) *Temperature*.—The temperatures of the flask and solution differ from the temperature at which the volume of the flask was calibrated. According to the manufacturer, the flask was calibrated at a temperature of 20°C, whereas the laboratory temperature varies by ±2°C. The uncertainty from this effect can be calculated from the estimate of the temperature range and the coefficient of the volume expansion. In the case of methanol (coefficient of volume thermal expansion, $\beta = 1.1 \cdot 10^{-3}$°C^{-1}) as the solvent, this effect is negligible.

Precision, A.—The uncertainty component was evaluated experimentally from the dispersion of repeated measurements. It can readily be expressed as the SD of the 10 measurements performed to determine A. Previously, the presence of outliers and the normal distribution of the data were evaluated by the 5% Dixon test and 5% Shapiro-Wilks standard test, respectively.

$$\left[\dot{u}_A\left(\overline{X}\right)\right]=SD\times\frac{1}{\sqrt{n}}\times\frac{1}{\overline{X}}$$

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Calculation of the combined standard uncertainty.—In order to calculate the relative combined standard uncertainty of a multiplicative expression (as above), the standard uncertainties of the components are used as follows:

$$\dot{u}\left(\overline{X}\right)=\sqrt{\begin{array}{l}\left[\dot{u}_A\left(\overline{X}\right)\right]^2+\left[\dot{u}_B\left(\overline{M}\right)\right]^2+\left[\dot{u}_B\left(\overline{C}x\right)\right]^2\\+\left[\dot{u}_B(V)\right]^2+\left[\dot{u}_B\left(\overline{E}\right)\right]^2+\left[\dot{u}_B(P)\right]^2\end{array}}$$

$\dot{u}_B\left(\overline{E}\right)$ was not considered as a component of the combined standard uncertainty when E did not significantly differ from 100%. The combined standard uncertainty is calculated as follows:

$$u\left(\overline{X}\right)=\dot{u}\left(\overline{X}\right)\cdot\overline{X}$$

The expanded uncertainty $U(r)$ is obtained by applying a coverage factor of 2 as follows:

$$U\left(\overline{X}\right)=2\cdot u\left(\overline{X}\right)$$

The relative combined standard uncertainty $\dot{u}(X)$, the absolute combined standard uncertainty, $u\left(\overline{X}\right)$, $\dot{u}\left(\overline{X}\right)$, and the expanded uncertainty, $U\left(\overline{X}\right)$, were calculated for all PGRs at 3 concentration levels. Moreover, the linearity of the squared combined standard uncertainty with the squared concentration level was calculated by using linear regression.

Determination of PGRs in commercial fertilizers.—Twenty samples of liquid fertilizers with declared biostimulant properties were collected in Italy in the framework of an investigation performed by the Italian Ministry of Agricultural Alimentary and Forestry Policies (MIPAAF) from January to April 2008. The pickup was made in 4 parts by MIPAAF officers from many Italian distributors, according to regulations (9, 10). An official report was prepared for every sample. The samples were labeled with a description of the pickup location and the pickup date and sent to the Catania Laboratory of the Central Inspectorate for Quality Control of Agricultural and Food Productions (ICQ) for the analysis. Once a sample was analyzed as described above, our laboratory issued a regular or irregular certificate of analysis.

Results and Discussion

Method Development for Determination of PGRs

The reversed-phase HPLC method described here was developed to provide a rapid quality control determination of PGRs in fertilizer. To obtain the best chromatographic conditions, the method was optimized by analysis of spiked fertilizers to provide sufficient selectivity and sensitivity in a short separation time. Different reversed C_{18} columns and mobile phases of acetonitrile–water or methanol–water were tested. Modifiers such as formic acid and acetic acid were tested. The best result was achieved with a Hydro-RP polar encapped C_{18} column, 250 × 4.60 mm (4 µm), with (0.8% acetic acid) water–acetonitrile (55 + 45) at a flow rate of 1.0 mL/min at 30°C. The use of acetonitrile as the organic component provided good sensitivity and a cleaner background. The water–acetonitrile proportions chosen provided better separation in a reasonable run time. For the selection of the best detection wavelength and for confirmation purposes, a photodiode array detector was used, and chromatograms were examined in the range of 200–360 nm. The 280 nm wavelength provided good detection of all compounds investigated, minimizing UV interferences detectable at lower wavelengths. Moreover, 280 nm is the λ_{max} of most of the PGRs investigated. Figure 1 shows the UV chromatograms (280 nm) of a spiked fertilizer with the 12 PGRs studied. The results showed that all the compounds eluted between 5 and 24 min. All peaks are reasonably resolved, with the exception of the pair MCPA and 2,4-D. Their separation should improve by using a higher percentage of water, but the longer run time would not be suitable for routine applications; moreover, it is improbable that a fertilizer additive would contain >1 PGR. For these reasons, the PGRs studied were divided into 2 separate mixtures. For the extraction solvent, pure methanol provided high recoveries in a short time and guaranteed complete dissolution of the compounds under study.

Method Validation

The method was fully validated according to International Union of Pure and Applied Chemistry (IUPAC) and EURACHEM guidelines (11, 12).

The specificity of the analytical method was demonstrated by evaluating the HPLC/UV chromatograms of 3 spiked and unspiked representative types of fertilizers such as liquid extracts from algae, liquid organic fertilizers, and liquid extracts from vegetal tissue containing amino acids and polysaccharides. Results demonstrated that other compounds in the sample extracts did not interfere at the detection wavelengths. Moreover, the RSD values for the retention times of the PGRs in the standard solutions and spiked fertilizer did not exceed 0.7% for all compounds.

The linearity of the detector response was determined by preparing calibration graphs. At 280 nm, the calibration curves of the PGRs were studied from about 0.5–25 mg/L (Table 2). The regression lines, with the corresponding coefficients of linear regression (r^2), slopes, and y-intercepts, calculated by the least-squares fit method, are also reported in Table 2. An acceptable coefficient of linear correlation must be >0.9900, and values from 0.996 to 0.999 for a 6-point calibration curve (4 replicates for each point) are considered proof of the linear relationship; moreover, the intercepts were not significantly different from zero for all investigated compounds.

Precision is usually expressed as the RSD of a series of measurements (13). In the study of the precision of the instrumental system, RSD values for measurement of peak areas ranged from 0.03% for NAD to 0.98% for thidiazuron (Table 2). Therefore, 1% could be considered the limit set for the precision of the instrumental system, which indicates that the equipment used for the study operated correctly for the developed method and produced highly repetitive results.

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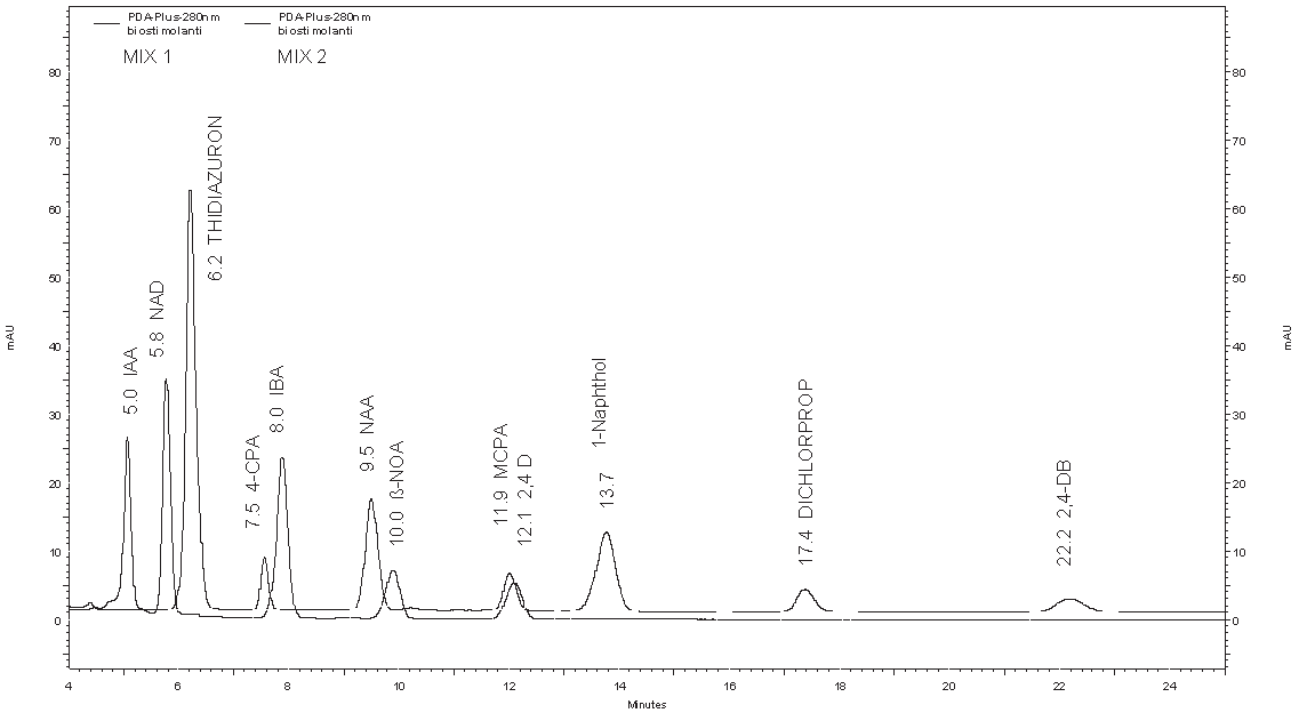

Figure 1. HPLC/UV chromatograms (280 nm) of a fortified fertilizer (concentrations of PGRs: 0.01%, w/w).

Table 2 also shows the instrumental detection and quantitation limits, expressed as mg/L, for all PGRs studied. The values were obtained by using the SD approach for analysis of a liquid organic fertilizer spiked with PGR concentrations close to the LOD values. The LOD is the lowest concentration of analyte that can be detected but not necessarily quantitated, whereas the LOQ represents the lowest concentration of analyte that can be quantitated with suitable precision and accuracy. The LOD values ranged from 0.08 mg/L for NAD to 0.27 mg/L for β-NOA, and the LOQ values ranged from 0.23 mg/L for NAD to 0.64 mg/L for 2,4-DB. Because LOD and LOQ values are parameters of qualitative and quantitative assays, respectively, the method LOD and LOQ values were calculated by considering the sample weight and the dilution procedure. The values obtained for all PGRs are reported in Table 2 and are expressed as %, w/w. The results showed method LOQ values <0.0064% for all PGRs; this LOQ value could be considered the overall method quantification value. Because the commercial products analyzed are concentrate formulations and must be diluted before use according to label directions, the LOQ values obtained were more than adequate to detect PGRs, which are effective at concentrations of 10^{-5}–10^{-6} M. For example, the LOQ value of 0.0025% for IAA corresponded to an active concentration of about $1.4 \cdot 10^{-7}$ M, whereas the LOQ value of 0.0064% for 2,4-DB reflected an active concentration of about $2.6 \cdot 10^{-7}$ M (representing a mean dilution of 100 g/100 L, as suggested for agricultural purposes). Nevertheless, more accurate and precise results could be obtained by starting with a larger sample; the ruggedness test (*see* related paragraph) demonstrated that the capacity of the method was not significantly affected by variation in sample weight.

The precision (repeatability) and accuracy (recovery) of the method were also demonstrated. The method repeatability, expressed as the RSD, was evaluated by analyzing 3 series of 10 fortified fertilizers at low, medium, and high concentrations (0.01, 0.05, and 0.1%, w/w; Table 3). The precision values were found to be <5.7% for all PGRs at medium and high concentration levels, whereas values as high as 6.2% were obtained for the lowest concentration of 0.01%, w/w. The best results were found for 4-CPA, which had an RSD of 2.7% at the highest concentration level (0.1%, w/w). The results discussed indicated good precision of the method at all concentration levels investigated, with no particular effect due to the concentration level and/or to a single PGR.

Recovery experiments were performed to determine the accuracy of the proposed method. Three different concentrations of PGRs were added to a negative control, and 10 independent replicates were analyzed for each concentration. Contrary to the results obtained for the precision study, the recovery data were affected by the concentration level considered, ranging from 79 to 104%. The mean recovery for all PGRs at all investigated concentrations was found to be 94% (SD, 7.5%); the mean recoveries were 100% (SD, 2.1%) for the highest concentration level, 96%

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Table 2. Validation parameters for the determination of PGRs

Parameter	IAA	NAD	Thidiazuron	4-CPA	IBA	NAA	β-NOA	MCPA	2,4-D	1-Naphthol	Dichlorprop	2,4-DB
Linearity range, mg/L	0.7–19.0	0.7–16.5	0.6–16.3	0.6–19.6	0.7–19.0	0.7–17.1	0.6–18.3	0.7–17.8	0.7–17.5	0.7–21.7	0.7–21.8	0.6–23.3
Slope[a]	136247	157326	357837	34784	198374	166222	67889	36254	31738	134339	29881	31560
y-Intercept[a]	–21856	–16546	17161	569	–27052	2932	–10680	–1166	–2894	5543	1638	–3636
r^2 [a,b]	0.9987	0.9960	0.9990	0.9993	0.9980	0.9992	0.9984	0.9960	0.9998	0.9990	0.9960	0.9991
RSD_{INSTR}, %[c]	0.52	0.03	0.98	0.92	0.46	0.76	0.71	0.60	0.04	0.96	0.40	0.90
LOD_{INSTR}, mg/L[d]	0.19	0.08	0.09	0.09	0.20	0.11	0.27	0.11	0.17	0.14	0.11	0.19
LOD_{METH}, %, w/w[e]	0.0019	0.0008	0.0009	0.0009	0.0020	0.0011	0.0027	0.0011	0.0017	0.0014	0.0011	0.0019
LOQ_{INSTR}, mg/L[f]	0.25	0.23	0.31	0.31	0.35	0.36	0.54	0.36	0.34	0.48	0.36	0.64
LOQ_{METH}, %, w/w[g]	0.0025	0.0023	0.0031	0.0031	0.0035	0.0036	0.0054	0.0036	0.0034	0.0048	0.0036	0.0064

[a] Values obtained from 6 independent stock solutions in 4 replicates.
[b] Correlation coefficient.
[c] Values obtained from 7 repeated scans of the same solution; RSD_{INSTR} = instrumental relative standard deviation.
[d] LOD_{INSTR} = Instrumental limit of detection.
[e] LOD_{METH} = Method limit of detection.
[f] LOQ_{INSTR} = Instrumental limit of quantification.
[g] LOQ_{METH} = Method limit of quantification.

(SD, 2.3%) for the intermediate concentration level, and 84% (SD, 4.2%) for the lowest concentration level (Table 3). The results of the accuracy test showed no significant difference among the investigated PGRs, with similar recoveries at each concentration level studied. The AOAC's Single Laboratory Validation document recommends general recovery limits of 75–120% at levels below 0.01%, w/w, and recovery limits of 80–115% at levels of 0.1%, w/w. The recovery data obtained are in accordance with the AOAC limits; recoveries were higher with respect to the AOAC guidelines for the 0.01% level. Therefore, the method should be considered extremely accurate.

The evaluation of robustness was considered during the development phase to show the reliability of the analysis with respect to deliberate variations in method parameters. The experiments were performed in duplicate at the 0.1%, w/w, concentration level by using the Youden plot for 2 variations (high and low values) of 7 selected parameters (Table 1); the results are reported in Table 4. The comparison between the ruggedness experiments/method repeatability data and *F*-test data showed the ruggedness of the method for all of the PGAs. All calculated SD_{Ru}^2/SD_M^2 ratios were appreciably below the *F*-value of 3.29. In the estimation of the effect of each parameter on method performance, the data showed positive results, demonstrating no particular effect due to a specific parameter. On the whole, the results demonstrated that measurements are not affected by variations in analytical conditions; therefore, no precautionary statement must be included in the procedure, and none of the analytical parameters investigated must be specially controlled. Only β-NOA and 4-CPA showed some outliers, indicating that the relative parameters (matrix effect and column temperature for β-NOA and sample weight for 4-CPA) must be well monitored during the application of the analytical protocol.

Results demonstrated, therefore, that the method could be suitable for PGR levels lower than those investigated, by starting from a larger amount of sample or a smaller volume of solvent and thereby improving the quantitative estimation.

Quantification of Uncertainty Sources

Uncertainty (of a result of measurement) is defined in the *Guide to the Expression of Uncertainty in Measurement* (GUM) as a parameter, associated with the result of measurement, that characterizes the dispersion of values that can be reasonably attributed to the measurand (14). The aim of this step is to quantify the uncertainty arising from each of the previously identified sources. This can be done by using experimental data or well-based assumptions.

The study was performed at 3 concentration levels (low, medium, and high), identifying and studying the most important parameters that determined the uncertainty of the analytical method.

The parameters selected were point calibration (Cx), mass (M), standard purity (P), recovery (E), volume (V), and precision (A); their contributions to method uncertainty were calculated as indicated in the *Experimental* section, according to the GUM procedure (14). The different contributions of

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Table 3. Repeatability and recovery of PGRs at 3 concentration levels

PGR	Spiked concn, %, w/w	Mean concn found, %, w/w[a]	RSD_{ID}, %[b]	E, %[c]
IAA	0.112	0.115	5.1	103
	0.056	0.054	4.1	96
	0.011	0.010	6.2	93
NAD	0.097	0.097	5.7	100
	0.049	0.048	4.0	98
	0.010	0.008	4.3	88
Thidiazuron	0.081	0.081	4.4	100
	0.041	0.039	4.0	96
	0.008	0.006	4.9	80
4-CPA	0.098	0.096	2.7	98
	0.049	0.046	4.1	93
	0.010	0.008	4.0	79
IBA	0.086	0.087	4.8	102
	0.043	0.042	5.2	98
	0.009	0.007	5.0	85
NAA	0.086	0.088	4.8	103
	0.043	0.043	5.2	99
	0.008	0.007	5.0	85
β-NOA	0.105	0.109	3.9	104
	0.053	0.052	4.3	100
	0.010	0.009	5.5	83
MCPA	0.099	0.097	5.7	98
	0.050	0.048	4.0	96
	0.010	0.009	4.3	86
2,4-D	0.108	0.107	5.6	99
	0.054	0.051	4.5	95
	0.011	0.009	4.1	82
1-naphthol	0.108	0.109	3.9	100
	0.054	0.052	4.3	96
	0.011	0.009	5.5	81
Dichlorprop	0.109	0.106	5.6	98
	0.055	0.051	4.5	94
	0.011	0.009	4.1	81
2,4-DB	0.116	0.116	5.1	99
	0.058	0.054	4.1	92
	0.012	0.010	6.2	89

[a] Each value is the mean of 10 experiments.
[b] RSD_{ID} = Intraday precision.
[c] E = Accuracy.

uncertainty for each concentration level together with the relative combined standard uncertainty, $u(\overline{X})$, are shown in Figure 2 for each PGR. Results showed that the contribution to uncertainty due to the dilution operations was constant for each concentration level, and that the contribution of standard purities was constant for each PGR. The same value of uncertainty concerning the amount of weighed sample was used for each level and for all PGRs, because the quantity of analyzed sample did not change among the experiments; moreover, this contribution could be considered negligible.

Regarding the 0.1%, w/w, level, Figure 2A shows that the purity of the standards is the largest contributor to measurement uncertainty for many PGRs, followed by repeatability. Only for 1-naphthol, with purity of 100%, was the main contribution to measurement uncertainty due to repeatability. On the contrary, the contribution of weight, dilution operations, and point calibration can be considered negligible. In fact, their removal from the formula yielding the standard uncertainty of measurement did not significantly change the final result.

Behavior was analogous for the 0.05%, w/w, level (Figure 2B), which showed that major contributors to the uncertainty of measurement were repeatability and standard purity. In some cases, the contributions due to recovery and calibration point were significant. At this level, the dilution operations represented a relatively considerable contribution.

In contrast, the 0.01%, w/w, level (Figure 2C) showed the calibration point as the main constituent of total uncertainty for all PGRs except NAD and IAA, for which all components contributed equally to measurement uncertainty.

When the uncertainty of the result is reported, the combined standard uncertainty is multiplied with a so-called coverage factor, yielding an expanded uncertainty (13). A factor $k = 2$ was used because of the resemblance of the expanded uncertainty to a 95% confidence interval. The values of combined standard and expanded uncertainty are shown in Table 5. Results showed a relative uncertainty (U%) ranging from 3 to 9% for all PGRs at levels of 0.1 and 0.05%, w/w. Higher values were obtained for the 0.01%, w/w, level. At this level, only NAD, β-NOA, and IAA had values of U% comparable with those obtained for the higher concentrations, whereas the maximum values of 21% were found for 1-naphthol, 2,4-DB, and 2,4-D.

Moreover, a linear regression could be applied to the squared combined standard uncertainty with the squared concentration level for all PGRs over the concentration range studied. Table 6 summarizes the linear regressions and correlation coefficients calculated for all PGRs. With the exception of 4-CPA, thidiazuron, and dichlorprop, all PGRs showed an excellent correlation (>0.99). Therefore, this relationship may be used to estimate the relative uncertainty of the PGRs for measured concentrations ±20% outside of the concentration range studied.

A comparison of the results obtained with the values calculated as described by Horwitz and Albert (15) for the same data set provided similar results for all PGRs (data not shown). The values of uncertainty of measurement estimated

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Table 4. Results of ruggedness testing

	IAA	NAD	Thidiazuron	4-CPA	IBA	NAA	β-NOA	MCPA	2,4-D	1-naphthol	Dichlorprop	2,4-DB
Mean detected concentration[a]	0.106	0.104	0.075	0.089	0.094	0.083	0.121	0.084	0.107	0.098	0.096	0.105
SD_{Ru}[b]	0.005	0.003	0.003	0.003	0.003	0.005	0.005	0.004	0.003	0.005	0.006	0.004
SD_{Ru}^2/SD_M^2[b]	0.84	0.27	0.59	1.71	0.59	1.67	1.18	0.48	0.31	1.27	0.86	0.37
$F_{v=7,9\ \alpha=0,05}$[c]	3.29	3.29	3.29	3.29	3.29	3.29	3.29	3.29	3.29	3.29	3.29	3.29
$\lvert V_{A/a}\rvert$(sample weight)[d]	0.0016	0.0029	0.0004	0.0044	0.0015	0.0028	0.0055	0.0015	0.0010	0.0037	0.0025	0.0005
$\lvert V_{B/b}\rvert$(solvent volume)[d]	0.0050	0.0026	0.0001	0.0002	0.0049	0.0012	0.0064	0.0018	0.0002	0.0025	0.0021	0.0030
$\lvert V_{C/c}\rvert$(sonication time)[d]	0.0021	0.0021	0.0008	0.0009	0.0002	0.0001	0.0061	0.0002	0.0058	0.0031	0.0019	0.0050
$\lvert V_{D/d}\rvert$(matrix type)[d]	0.0016	0.0030	0.0019	0.0006	0.0003	0.0063	0.0113	0.0028	0.0004	0.0036	0.0046	0.0015
$\lvert V_{E/e}\rvert$(solvent composition)[d]	0.0074	0.0002	0.0042	0.0028	0.0002	0.0026	0.0063	0.0004	0.0018	0.0000	0.0009	0.0002
$\lvert V_{F/f}\rvert$(ultrasonic bath temperature)[d]	0.0003	0.0005	0.0021	0.0029	0.0030	0.0055	0.0008	0.0034	0.0011	0.0038	0.0043	0.0013
$\lvert V_{G/g}\rvert$(column temperature)[d]	0.0033	0.0004	0.0008	0.0019	0.0010	0.0042	0.0090	0.0052	0.0003	0.0047	0.0073	0.0026
$t_{P=0.95\ v=9}\ SD_M/\sqrt{2}$[e]	0.0093	0.0089	0.0058	0.0042	0.0067	0.0067	0.0067	0.0089	0.0096	0.0067	0.0096	0.0094

[a] $n = 8$; 2 Replicates for each experiment; spike of about 0.1%, w/w, of each PGR.
[b] SD_{Ru} = Ruggedness test standard deviation; SD_M = method standard deviation (spike concentration, 0.1%, w/w; $n = 10$).
[c] F = F-test; v = degrees of freedom; α = level of significance.
[d] $V_{X/x}$ = Effect of X and x; X = parameter considered.
[e] t = Student's t value; P = level of confidence.

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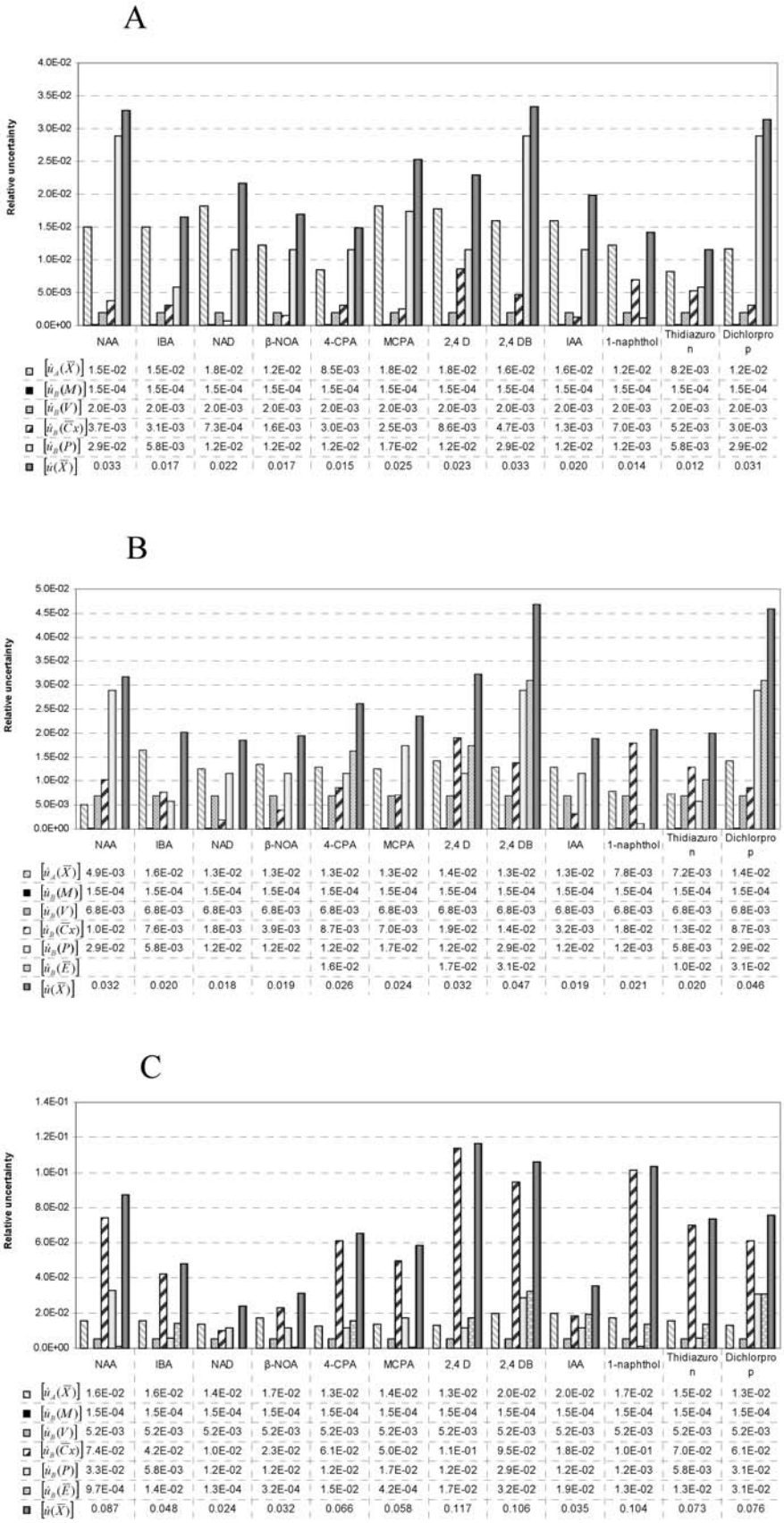

Figure 2. Uncertainty contributions and relative combined standard uncertainty values of PGRs, calculated for concentration levels of (A) 0.1%, w/w; (B) 0.05%, w/w; and (C) 0.01%, w/w.

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Table 5. Uncertainty components for PGRs at 3 concentration levels PGR

PGR	$\overline{X}$[a]	u($\overline{X}$)[b]	U($\overline{X}$)[c]	U(%)[d]
		0.1%, w/w		
IAA	0.116	0.0023	0.0046	4
NAD	0.097	0.0021	0.0042	4
Thidiazuron	0.081	0.0009	0.0019	2
4-CPA	0.096	0.0014	0.0029	3
IBA	0.088	0.0015	0.0029	3
NAA	0.088	0.0029	0.0058	7
β-NOA	0.109	0.0018	0.0037	3
MCPA	0.097	0.0025	0.0049	5
2,4-D	0.107	0.0025	0.0049	5
1-Naphthol	0.109	0.0016	0.0031	3
Dichlorprop	0.108	0.0034	0.0068	6
2,4-DB	0.116	0.0039	0.0077	7
		0.05%, w/w		
IAA	0.054	0.0010	0.0020	4
NAD	0.048	0.0009	0.0018	4
Thidiazuron	0.039	0.0008	0.0016	4
4-CPA	0.046	0.0012	0.0024	5
IBA	0.042	0.0009	0.0017	4
NAA	0.043	0.0014	0.0027	6
β-NOA	0.052	0.0010	0.0020	4
MCPA	0.048	0.0011	0.0022	5
2,4-D	0.051	0.0017	0.0033	6
1-Naphthol	0.053	0.0011	0.002	4
Dichlorprop	0.051	0.0023	0.0047	9
2,4-DB	0.054	0.0025	0.0050	9
		0.01%, w/w		
IAA	0.010	0.0004	0.0007	7
NAD	0.008	0.0002	0.0004	5
Thidiazuron	0.006	0.0005	0.0009	15
4-CPA	0.008	0.0005	0.0010	13
IBA	0.007	0.0004	0.0007	10
NAA	0.007	0.0006	0.0013	17
β-NOA	0.009	0.0003	0.0006	6
MCPA	0.008	0.0005	0.0010	12
2,4-D	0.009	0.0011	0.0022	21
1-Naphthol	0.009	0.0009	0.0018	21
Dichlorprop	0.009	0.0007	0.0013	15
2,4-DB	0.010	0.0011	0.0022	21

[a] %, w/w = Mean concentration found; n = 10.
[b] Combined standard uncertainty.
[c] Expanded uncertainty (k = 2).
[d] Relative expanded uncertainty.

Table 6. Linearity of the squared combined standard uncertainty $u(\overline{X})$ with the squared concentration level in the range of 0.01–0.1%, w/w

PGR	m[a]	q[b]	r^{2c}
IAA	3.9E-04	6.6E-09	0.9988
NAD	4.8E-04	–1.2E-07	0.9957
Thidiazuron	7.3E-05	3.7E-07	0.8267
4-CPA	1.6E-04	6.2E-07	0.7775
IBA	2.6E-04	2.3E-07	0.9910
NAA	1.1E-03	1.8E-07	0.9987
β-NOA	2.6E-04	1.7E-07	0.9949
MCPA	6.6E-04	–1.4E-08	0.9933
2,4-D	4.3E-04	1.4E-06	0.9855
1-Naphthol	1.5E-04	8.0E-07	1.0000
Dichlorprop	8.9E-04	1.5E-06	0.9441
2,4-DB	1.1E-03	5.0E-07	0.9917

[a] Slope.
[b] Intercept.
[c] Correlation coefficient.

in this study should be considered very good and adequate for the concentration levels studied. In a different approach to the uncertainty expression, the application of a stepwise implementation of the uncertainty concept allowed a quantitative estimation of the different source of errors in the measurand evaluation, at various concentration levels. This kind of information helps the development of the analytical protocol with the aim of improving the method itself.

Determination of PGRs in Commercial Fertilizers

The methodology discussed above was used to investigate the occurrence of PGRs in 20 samples of commercial biostimulant fertilizers, collected in Italy as described in the *Experimental* section. The presence of NAA was found in 4 of the 20 samples analyzed in the following concentrations: 0.011, 0.031, 0.033, and 0.46%, w/w. These results demonstrated the efficacy of the extraction protocol with real matrixes beyond the laboratory-fortified samples, and the amounts found were at least 3 times higher than the LOQ value for this PGR. These data are in agreement with the previously discussed active concentrations of these substances, based on users' suggestions for agricultural applications (dilutions of about 1–100).

In conclusion, a rapid and reliable isocratic HPLC/PDA method has been developed for the determination of PGRs in fertilizers. The proposed reversed-phase LC method is precise, accurate, and specific, without significant matrix effects. The method was completely validated, showing satisfactory data for all method validation parameters tested. The study of the uncertainty of measurements revealed the relative significance of monitoring selected parameters during method application in order to minimize sources of

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uncertainty in quantitative results. The developed method can be used for the routine analysis of fertilizers with declared biostimulant properties in order to avoid the use of PGRs.

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